

Name: Index No.

School: Signature:

553/2
BIOLOGY
PAPER 2
July/August 2025
2½ hours



WAKISSHA JOINT MOCK EXAMINATIONS

Uganda Certificate of Education

BIOLOGY

Paper 2

Practical

2 hours 30 minutes

INSTRUCTIONS TO CANDIDATES:

This paper consists of two examination items. Answer all the items in the spaces provided.

Drawings should be made in the spaces provided. Use sharp pencils for your drawings. Coloured pencils or crayons should not be used.

No additional sheets of writing paper are to be inserted in the booklet.

Work on additional sheets will not be scored.

FOR SCORERS' USE ONLY

Item 1

Basic code	A	V	H	C	D	AR	Total score	Scorer's initials
Score								

Item 2

Basic code	I	S	DR	Total score	Scorer's initials
Score					

Item 1

James and Simon are farmers in Kiyiki village, who planted Irish potatoes at the same time during the last planting season.

They both applied the same type of fertilizer.

James used the recommended concentration and his plants grew healthy and strong, producing a good harvest.

However, Simon used twice the recommended concentration, thinking that it would boost the yields.

Unfortunately at flowering stage, most of his plants withered and the harvest was poor.

You are provided with the two solutions A and B which are of different concentrations of the fertilizer used by the two farmers and specimen Q which is a plant part obtained from the plants grown by the two farmers.

Task:

Design and carryout a scientific investigation on solutions A and B using Specimen Q and use your results to explain why Simon had a poor harvest and advise the farmers on the correct concentration of the fertilizer to be used.

Your design and investigation should include the following;

- (a) (i) Aim, Hypothesis and variables

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There is no handwriting or printed text on the paper. A small dark mark is visible near the top right corner.

Item 2

Mr. Okello, a farmer in Lira grows a number of crops in his garden, including beans and maize. In the middle of the garden he grows crops with broad net veined leaves and grows plants with narrow parallel veined leaves at the edge of the garden. You are provided with specimens R, S and T obtained from Okello's garden.

Task

- (a) With reasons, identify the plant organs from the specimens that were obtained from Okello's garden.

- (b) Explain how the specimens that were obtained from the middle of the garden are adapted for their functions.

- (c) Draw and label the main parts of a specimen that was obtained from the edge of Okello's garden.

Each candidate should be provided with;

- 20 cm³ of 1.0 M sucrose solution, labelled A
- 20 cm³ of distilled water, labelled B
- A medium size Irish potato, labelled Q
- A cork borer of diameter between 0.5 cm to 1.0 cm.
- A knife
- 2 test tubes
- Labels
- Stop clock
- A bean leaf, labelled R
- A maize leaf, labelled S
- A sweet potato leaf, labelled T
- 10 ml measuring cylinder.